

19 - Nitrogen compounds

05/01/25

Amines

∴ production ⇒ reaction of a halogenoalkane with NH_3 in ethanol * [conditions: heated under pressure] *



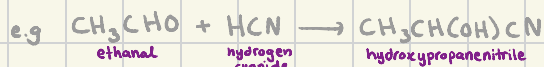
Nitriles

∴ production ⇒ reaction of a halogenoalkane with KCN in ethanol * [conditions: heated] *



hydroxynitriles

∴ production ⇒ reaction of an aldehyde / ketone with HCN * [conditions: KCN as a catalyst & heat] *



hydrolysis of nitriles

dilute acid or dilute alkali & heat, followed by acidification (if using alkali)

products: carboxylic acid & ammonium salt (with acid hydrolysis) or carboxylate salt (with alkaline hydrolysis)

